

In addition to its technical characteristics, SM6 meets requirements concerning safety of life and property as well as ease of installation, operation and protecting the environment.

SM6 units are designed for indoor installations.

Their compact dimensions are:

- 375 to 1500 mm width
- 1600 to 2250 mm height
- 840 to 1400 mm depth...

... this makes for easy installation in small rooms or prefabricated substations. Cables are connected via the front.

All control functions are centralised on a front plate, thus simplifying operation. The units may be equipped with a number of accessories (relays, toroids, instrument transformers, surge arrester, control and monitoring, etc.).

Normal operating conditions

- **Ambient air temperature:**

- 1) less than or equal to 40°C
- 2) less than or equal to 35°C on average over 24 hours
- 3) greater or equal to -5°C.

- **Altitude**

- 1) less than or equal to 1000 m
- 2) above 1000 m, a derating coefficient is applied (please consult us).

- **Solar radiation**

- 1) no solar radiation influence is permitted.

- **Ambient air pollution**

- 1) no significant pollution by dust, smoke, corrosive and/or flammable gases, vapours or salt.

- **Humidity**

- 1) average relative humidity over a 24 hour period, less than or equal to 95%
- 2) average relative humidity over a 1 month period, less than or equal to 90%
- 3) average vapor pressure over a 24 hour period, less than or equal to 2.2 kPa
- 4) average vapor pressure over a 1 month period, less than or equal to 1.8 kPa.

For these conditions, condensation may occasionally occur. Condensation can be expected where sudden temperature changes occur in periods of high humidity.

To withstand the effects of high humidity and condensation, such as breakdown of insulation, please pay attention on Civil Engineering recommendations for design of the building or housing, by suitable ventilation and installation.

- **Seismic (for 24 Kv and 36 kV):**

- 1) Up to 0.5 g (horizontal) and 0.4 g (vertical)
- 2) Class 2 for 24 kV and Class 1 for 36kV
- 3) According to standards IEEE-693/2005 and EN 60068-3/1993 for the 24 kV and 36 kV

Severe operating conditions (please consult us).

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SM6 units meet all the following standards and specifications:

- IEC standards
- UTE standards for SM6-24
- EDF specifications for SM6-24
- SEISMIC standards for 24 kV

IEC standards

62271-200	High-voltage switchgear and controlgear - Part 200: A.C. metal-enclosed switchgear and controlgear for rated voltage above 1 kV and up to and including 52 kV.
62271-1	High-voltage switchgear and controlgear - Part 1: Common specifications.
62271-103	High voltage switches - Part 1: switches for rated voltages above 1 kV and less or equal to 52 kV.
62271-105	High-voltage switchgear and controlgear - Part 105: High voltage alternating current switch-fuse combinations.
60255	Electrical relays.
62271-100	High-voltage switchgear and controlgear - Part 100: High-voltage alternating current circuit breakers.
62271-102	High-voltage switchgear and controlgear - Part 102: High-voltage alternating current disconnectors and earthing switches.
61869-2	Instrument transformers - Part 1: Current transformers.
61869-3	Instrument transformers - Part 2: Voltage transformers.
60044-8	Instrument transformers - Part 8: Low Power Current Transducers.
62271-206	High-voltage prefabricated switchgear and controlgear assemblies - Voltage presence indicating systems.
62271-304	High-voltage switchgear and controlgear - Part 304: Design classes for indoor enclosed switchgear and controlgear for rated voltages above 1 kV up to and including 52 kV to be used in severe climatic conditions.

SEISMIC standards for 24kV

IEE-693	2005 IEEE Recommended Practice for Seismic Design of Substations
EN600068-3-3	1993 Environmental testing-Part 3: guidance, Seismic test methods for equipments

UTE standards for 24 kV

NFC 13.100	Consumer substation installed inside a building and fed by a second category voltage public distribution system.
NFC 13.200	High voltage electrical installations requirements.
NFC 64.130	High voltage switches for rated voltage above 1 kV and less than 52 kV.
NFC 64.160.	Alternating current disconnectors and earthing switches



The hereunder values are for working temperatures from -5°C up to +40°C and for a setting up at an altitude below 1000 m.

Electrical characteristics

Rated voltage	Ur	kV		7.2	12	17.5	24	36
Insulation level								
Insulation	Ud	50/60 Hz, 1 min (kV rms)		20	28	38	50	70
Isolation	Ud	50/60 Hz, 1 min (kV rms)		23	32	45	60	80
Insulation	Up	1.2/50 μs (kV peak)		60	75	95	125	170
Isolation	Up	1.2/50 μs (kV peak)		70	85	110	145	195
Breaking capacity								
Transformer off load		A		16				
Cables off load		A		31.5				50
Rated current	Ir	A		400 - 630 -1250				630-1250
Short-time withstand current	Ik/tk ⁽¹⁾	kA/1 s	25	630 - 1250				1250
			20 ⁽²⁾	630 - 1250				
			16	630 - 1250				
			12.5	400 - 630 - 1250				630-1250
			62.5	630				
Making capacity (50 Hz)	Ima	kA	50	630				
			40	630				
			31.25	400 - 630				630
Maximum breaking capacity (Isc)								
Units IM, IMC, IMB, IMM ⁽⁴⁾		A		630 - 800 ⁽³⁾				630
NSM-cables, NSM-busbars		A		630 - 800 ⁽³⁾				NA
QM, QMC, QMB		kA		25		20		20
PM		kA		25				20
CVM		kA		6.3	NA			
CVM with fuses		kA		25	NA			
SF6 circuit breaker range								
DM1-A, DM1-D, DM1-W ⁽⁴⁾ , DM1-M ⁽⁴⁾	kA	25		630-1250				1250
			20	630-1250				
DM1-S	kA	25		630				NA
DM1-Z		25		1250				NA
DM2	kA	20		630				
			25	630				1250
Vacuum circuit breaker range								
DMV-A, DMV-D	kA	25		630-1250				NA
DMVL-A	kA	20		630				NA
DMVL-D	kA	25		630				NA

NA: Non Available

(1) 3 phases

(2) In 20 kA/3 s for SM6-24 only, consult us

(3) In 800 A, consult us.

(4) NA for SM6-36

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**Protection index:**

- Classes: PI (insulating partition)
- Loss of service continuity classes: LSC2A (LSC1 for metering GAM/GBM functions)
- Units in switchboard: IP3X
- Between compartments: IP2X for SM6-24, IP2XC for SM6-36
- Cubicle: IK08 for SM6-24, IK07 for SM6-36.

Electro-magnetic compatibility:

- Relays: 4 kV withstand capacity, as per recommendation IEC 60801.4
- Compartments:
 - electrical field:
 - 40 dB attenuation at 100 MHz
 - 20 dB attenuation at 200 MHz
 - magnetic field: 20 dB attenuation below 30 MHz.
- 3) According to standards IEEE-693/2005 and EN 60068-3/1993
- **for 36 kV** (please contact us).

Temperatures:

The cubicles must be stored and installed in a dry area free from dust and with limited temperature variations.

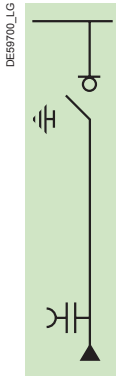
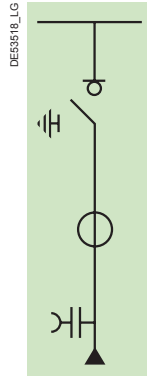
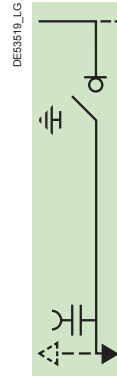
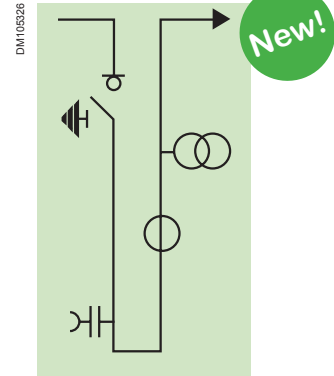
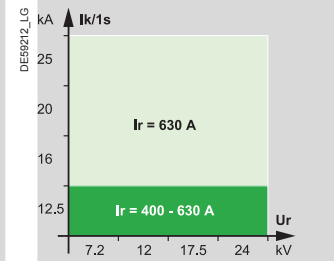
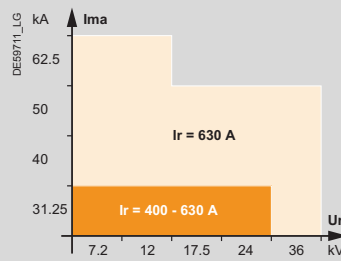
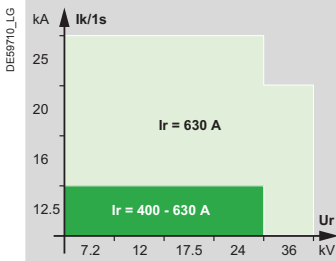
- For stocking: from -40°C to +70°C
- For working: from -5°C to +40°C
- Other temperatures, consult us.

Seismic:

- **for 24 kV** (option):
 - 1) Up to 0.5 g (horizontal) and 0.4 g (vertical)
 - 2) Class 2

Endurance

Units		Mechanical endurance	Electrical endurance
IM, IMC, IMB, IMM, PM, QM ⁽⁵⁾ , QMC ⁽⁵⁾ , QMB ⁽⁵⁾ , NSM-cables, NSM-busbars		IEC 62271-103 1 000 operations class M1	IEC 62271-103 100 breaks at Ir, p.f. = 0.7, class E3
CVM	Disconnector	IEC 62271-102 1 000 operations	
	Vacuum contactor	IEC 60470 2 500 000 operations 250 000 with mechanical latching	IEC 60470 250 000 breaks at Ir
SF6 circuit breaker range			
DM1-A, DM1-D, DM1-M, DM1-W, DM1-Z, DM1-S, DM2	Disconnector	IEC 62271-102 1 000 operations	
	SF circuit breaker	IEC 62271-100 10 000 operations class M2	IEC 62271-100 30 breaks at 12.5 kA for SM6-24 25 breaks at 25 kA for SM6-24 40 breaks at 16 kA for SM6-36 15 breaks at 25 kA for SM6-36 10 000 breaks at Ir, p.f. = 0.7, class E2
		Operating sequence	O - 0.3 s - CO - 15 s - CO O - 0.3 s - CO - 3 mn O - 3 mn - CO - 3 mn - CO
Vacuum circuit breaker range			
DMV-A, DMV-D	Switch	IEC 62271-103 1 000 operations class M1	IEC 62271-103 100 breaks at Ir, p.f. = 0.7, class E3
	Evolis circuit breaker	IEC 62271-100 10 000 operations class M2	IEC 62271-100 100 breaks at 25kA for SM6-24 10 000 breaks at Ir, p.f. = 0.7, class E2
DMVL-A, DMVL-D	Disconnector	IEC 62271-102 1 000 operations	
	Evolis circuit breaker	IEC 62271-100 10 000 operations class M2	IEC 62271-100 100 breaks at 16kA for SM6-24 100 breaks at 25kA for SM6-24 10 000 breaks at Ir, p.f. = 0.7, class E2
(5) As per recommendation IEC 62271-105, three breakings at p.f. = 0.2 800 A under 36 kV; 1400 A under 24 kV; 1730 A under 12 kV; 2600 A under 5.5 kV.			
Internal arc withstand (in accordance with IEC 62271-200):			
• SM6-24:			
Basic		□ 12.5 kA 1 s, IAC: A-FL	
Advance		□ 12.5 kA 1 s, IAC: A-FLR	
		□ 16 kA 1 s, IAC: A-FLR & IAC: A-FL	
		□ 20 kA 1 s, IAC: A-FLR & IAC: A-FL	
• SM6-36:			
		□ 16 kA 1 s, IAC: A-FL	

IM
Switch unit**IMC**
Switch unit**IMB**
Switch unit **with** earthing
switch, right or left outgoing**IMM**
Switch and measurement unit,
right or left outgoing line**Electrical characteristics****Basic equipment:**

- switch and earthing switch
- three-phase busbars
- CIT operating mechanism

- voltage presence indicator
- 150 W heating element for SM6-36
- LSC2A

- connection pads for dry-type cables

- three-phase bottom busbars for outgoing lines (right or left)

- one to three CT's for SM6-24
- three CT's for SM6-36

- Three CT's

Versions:

- CI2 operating mechanism

- CI1 operating mechanism

- CI1 operating mechanism for SM6-36

- CI1 operating mechanism

- in 800 A version for SM6-24, consult us

Option:

- motor for operating mechanism
- motor with severe and communication
- auxiliary contacts
- key-type interlocks
- release units (coil)
- operation counter
- 1250 A three-phase upper busbars
- 630 A three-phase upper busbars for severe operating conditions for SM6-24
- visibility of main contacts
- pressure indicator device
- enlarged low-voltage control cabinet for SM6-24
- 50 W heating element for SM6-24

- earth fault indicators
- connection pads for two dry-type single-core cables for 36 kV

- digital ammeter
- surge arresters (for SM6-36 and for SM6-24 in 500 mm width cubicle)
- 630 A busbars earthing switch cabinet for SM6-24 (not available for internal arc IEC62271-200)
- arc detection
- thermal monitoring
- ambient monitoring (only IM cubicle)

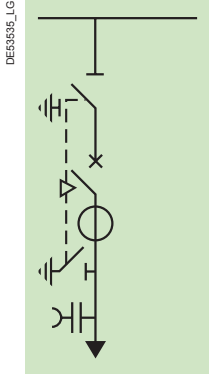
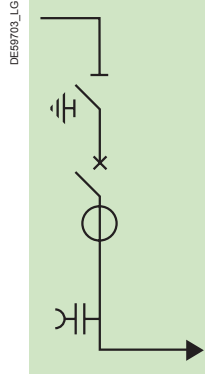
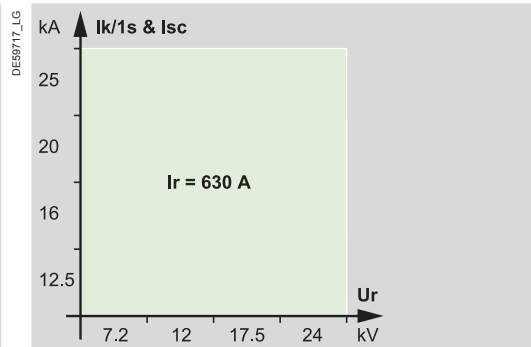
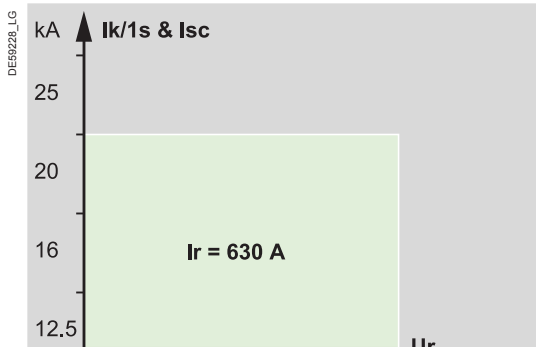
- arc detection
- thermal monitoring

- 630 A cable connection by the top (no internal arc withstand if selected)
- protection using Sepam programmable electronic unit
- three voltage transformers
- key-type interlocks

- arc detection

DMVL-A

Single-isolation disconnectable circuit breaker unit

**DMVL-D**Single-isolation disconnectable circuit breaker unit
Outgoing line on right**Electrical characteristics****Basic equipment:**

- Evolis circuit breaker lateral disconnectable
- disconnecter and earthing switch
- mechanical interlocking between circuit breaker and disconnecter
- three-phase busbars
- circuit breaker operating mechanism RI
- disconnecter operating mechanism CS
- voltage presence indicator
- auxiliary contacts on circuit breaker
- 3 CTs
- connection pads for dry-type cables
- LSC2A

- downstream earthing switch 2 kA rms making capacity

Option:

- **cubicle:**
 - auxiliary contacts on the disconnecter
 - three voltage transformers
 - key-type interlocks
 - 50 W heating element
 - 1250 A three-phase upper busbars at $I_r 630 \text{ A}$
 - 630 A three-phase upper busbars for severe operating conditions
 - enlarged low-voltage control cabinet
 - Sepam relay protection
 - Easergy P3 relay
 - surge arresters
 - thermal monitoring
 - arc detection
- **circuit breaker:**
 - motor for operating mechanism
 - release units (coil)
 - operation counter on manual operating mechanism